

Electric Charges and Fields

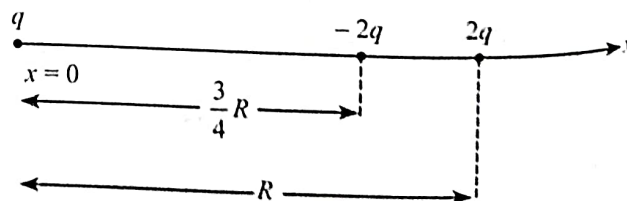
Coulomb's Law

1. A positive ion A and a negative ion B has charges 6.67×10^{-19} C and 9.6×10^{-10} C, and masses 19.2×10^{-27} kg and 9×10^{-27} kg respectively. At an instant, the ions are separated by a certain distance r . At that instant the ratio of the magnitudes of electrostatic force to gravitational force is $P \times 10^{-13}$, where the value of P is _____.
[23 Jan, 2025 (Shift-I)]
(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-1}$ and universal gravitational constant as $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$)
2. A small uncharged conducting sphere is placed in contact with an identical sphere but having 4×10^{-8} C charge and then removed to a distance such that the force of repulsion between them is 9×10^{-3} N. The distance between them is (Take $\frac{1}{4\pi\epsilon_0}$ as 9×10^9 in SI units)
[24 Jan, 2025 (Shift-II)]
(a) 3 cm (b) 1 cm (c) 4 cm (d) 2 cm
3. Force between two point charges q_1 and q_2 placed in vacuum at ' r ' cm apart is F . Force between them when placed in a medium having dielectric $K = 5$ at ' $r/5$ ' cm apart will be: [31 Jan, 2024 (Shift-II)]
(a) $F/25$ (b) $5F$ (c) $F/5$ (d) $25F$
4. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is 1.5 g/cc, the dielectric constant of water will be _____
5. Two identical charged spheres are suspended by string of equal lengths. The string make an angle of 37° with each other. When suspended in a liquid of density 0.7 g/cm^3 , the angle remains same. If density of material of the sphere is 1.4 g/cm^3 , the dielectric constant of the liquid is _____. ($\tan 37^\circ = \frac{3}{4}$) [30 Jan, 2024 (Shift-II)]
6. A thin metallic wire having cross sectional area of 10^{-4} m^2 is used to make a ring of radius 30 cm. A positive charge of 2π C is uniformly distributed over the ring, while another positive charge of 30 pC is kept at the centre of the ring. The tension in the ring is _____ N; provided that the ring does not get deformed (neglect the influence of gravity).

(given $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI units}$)

[27 Jan, 2024 (Shift-I)]

7. Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4 \text{ N/C}$ normally. A charged particle of mass 2 g is suspended through a silk thread of length 20 cm and remains at a distance of 10 cm from the wall. Then the charge on the particle will be $\frac{1}{\sqrt{x}} \mu\text{C}$ where $x = \underline{\hspace{2cm}}$. [use $g = 10 \text{ m/s}^2$]
[1 Feb, 2024 (Shift-I)]
8. In hydrogen like system the ratio of coulombian force to gravitational force between an electron and a proton is in the order of:
[05 April, 2024 (Shift-I)]
(a) 10^{39} (b) 10^{19} (c) 10^{29} (d) 10^{36}
9. Two identical conducting spheres P and S with charge Q on them repel each other with a force 16 N . A third identical uncharged conducting sphere R is successively brought in contact with the spheres. The new force of repulsion between P and S is:
[06 April, 2024 (Shift-I)]
(a) 4 N (b) 6 N (c) 1 N (d) 12 N
10. If two charges q_1 and q_2 are separated with distance ' d ' and placed in a medium of dielectric constant K . What will be the equivalent distance between charges in air for the same electrostatic force?
[24 Jan, 2023 (Shift-I)]
(a) $d\sqrt{k}$ (b) $k\sqrt{d}$ (c) $1.5d\sqrt{k}$ (d) $2d\sqrt{k}$
11. A $10 \mu\text{C}$ charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are:
[13 April, 2023 (Shift-I)]
(a) $9 \mu\text{C}$, $1 \mu\text{C}$ (b) $5 \mu\text{C}$, $5 \mu\text{C}$
(c) $7 \mu\text{C}$, $3 \mu\text{C}$ (d) $8 \mu\text{C}$, $2 \mu\text{C}$
12. Three point charges q , $-2q$ and $2q$ are placed on x-axis at a distance $x=0$, $x=\frac{3}{4}R$ and $x=R$ respectively from origin as shown. If $q = 2 \times 10^{-6} \text{ C}$ and $R = 2 \text{ cm}$, the magnitude of net force experienced by the charge $-2q$ is _____ N. [13 April, 2023 (Shift-I)]

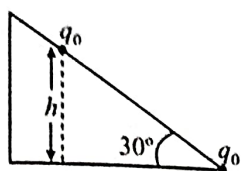


13. As shown in the figure, a configuration of two equal point charges ($q_0 = +2 \mu\text{C}$) is placed on an inclined plane. Mass of each point charge is 20 g . Assume that there is no friction between charge and the plane.

plane. For the system of two point charges to be in equilibrium (at rest) the height $h = x \times 10^{-3}$ m. The value of x is _____ mm.

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$, $g = 10 \text{ ms}^{-2}$)

[11 April, 2023 (Shift-I)]



14. Two equal positive point charges are separated by a distance $2a$. The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes maximum is $\frac{a}{\sqrt{x}}$.

The value of x is

[1 Feb, 2023 (Shift-I)]

15. A point charge $q_1 = 4q_0$ is placed at origin. Another point charge $q_2 = -q_0$ is placed at $x = 12 \text{ cm}$. Charge of proton is q_0 . The proton is placed on x -axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is _____ cm.

[29 Jan, 2023 (Shift-I)]

16. Two identical positive charges Q each are fixed at a distance of ' $2a$ ' apart from each other. Another point charge q_0 with mass ' m ' is placed at midpoint between two fixed charges. For a small displacement along the line joining the fixed charges, the charge q_0 executes SHM. The time period of oscillation of charge q_0 will be:

[27 July, 2022 (Shift-I)]

- (a) $\sqrt{\frac{4\pi^3\epsilon_0 m a^3}{q_0 Q}}$ (b) $\sqrt{\frac{q_0 Q}{4\pi^3\epsilon_0 m a^3}}$
(c) $\sqrt{\frac{2\pi^2\epsilon_0 m a^3}{q_0 Q}}$ (d) $\sqrt{\frac{8\pi^3\epsilon_0 m a^3}{q_0 Q}}$

17. Two identical metallic spheres A and B when placed at certain distance in air repel each other with force of F . Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between sphere A and B . The force experienced by sphere C will be:

[29 July, 2022 (Shift-II)]

- (a) $3F/2$ (b) $3F/4$ (c) F (d) $2F$

18. Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will experience the maximum Coulomb's force is:

[29 June, 2022 (Shift-II)]

- (a) $x = d$ (b) $x = \frac{d}{2}$ (c) $x = \frac{d}{\sqrt{2}}$ (d) $x = \frac{d}{2\sqrt{2}}$

19. Two identical charged particles each having a mass 10 g and charge $2.0 \times 10^{-7} \text{ C}$ are placed on a horizontal table with a separation of L between them such that they stay in limited equilibrium. If the coefficient of friction between each particle and the table is 0.25 , find the value of L .

[24 June, 2022 (Shift-II)]

- (a) 12 cm (b) 10 cm (c) 8 cm (d) 5 cm

20. A charge of $4\mu\text{C}$ is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be:

[27 July, 2022 (Shift-II)]

(a) $1\mu\text{C}$ and $3\mu\text{C}$

(b) $2\mu\text{C}$ and $2\mu\text{C}$

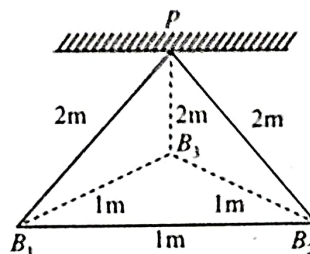
(c) 0 and $4\mu\text{C}$

(d) $1.5\mu\text{C}$ and $2.5\mu\text{C}$

21. Three identical charged balls each of charge 2C are suspended from a common point P by silk threads of 2 m each (as shown in figure). They form an equilateral triangle of side 1 m .

The ratio of net force on a charged ball to the force between any two charged balls will be

[27 June, 2022 (Shift-II)]



- (a) $1:1$ (b) $1:4$ (c) $\sqrt{3}:2$ (d) $\sqrt{3}:1$

22. Three point charges of magnitude $5\mu\text{C}$, $0.16\mu\text{C}$, and $0.3\mu\text{C}$ are located at the vertices A , B , C of a right angled triangle whose sides are $AB = 3 \text{ cm}$, $BC = 3\sqrt{2} \text{ cm}$ and $CA = 3 \text{ cm}$ and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

[26 July, 2022 (Shift-II)]

23. Two identical tennis balls each having mass ' m ' and charge ' q ' are suspended from a fixed point by threads of length ' l '. What is the equilibrium separation when each thread makes a small angle ' θ ' with the vertical?

[27 July, 2021 (Shift-I)]

- (a) $x = \left(\frac{q^2 l}{2\pi\epsilon_0 m g} \right)^{1/2}$ (b) $x = \left(\frac{q^2 l}{2\pi\epsilon_0 m g} \right)^{1/3}$
(c) $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g^2} \right)^{1/3}$ (d) $x = \left(\frac{q^2 l^2}{2\pi\epsilon_0 m^2 g} \right)^{1/3}$

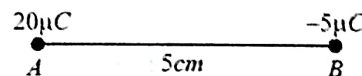
24. A certain charge Q is divided into parts q and $(Q - q)$. How should the charges Q and q be divided so that q and $(Q - q)$ placed at a certain distance apart experience maximum electrostatic repulsion?

[20 July, 2021 (Shift-I)]

- (a) $Q = \frac{q}{2}$ (b) $Q = 4q$ (c) $Q = 2q$ (d) $Q = 3q$

25. Two particles A and B having charges $20 \mu\text{C}$ and $-5 \mu\text{C}$ respectively are held fixed with a separation of 5 cm . At what position a third charged particle should be placed so that it does not experience a net electric force?

[31 Aug, 2021 (Shift-I)]



- (a) At 5 cm from $20 \mu\text{C}$ on the left side of system.
(b) At 1.25 cm from $-5 \mu\text{C}$ between two charges.
(c) At midpoint between two charges.
(d) At 5 cm from $-5 \mu\text{C}$ on the right side.

26. Two electrons each are fixed at a distance ' $2d$ '. A third charge proton placed at the midpoint is displaced slightly by a distance x ($x \ll d$) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency:

(m = mass of charged particle) [24 Feb, 2021 (Shift-II)]

- (a) $\left(\frac{2q^2}{\pi\epsilon_0 m d^3} \right)^{1/2}$ (b) $\left(\frac{2\pi\epsilon_0 m d^3}{q^2} \right)^{1/2}$
(c) $\left(\frac{q^2}{2\pi\epsilon_0 m d^3} \right)^{1/2}$ (d) $\left(\frac{2\pi\epsilon_0 m d^3}{2q^2} \right)^{1/2}$

27. Two small spheres each of mass 10 mg are suspended from a point by threads 0.5 m long. They are equally charged and repel each other to a distance of 0.20 m. Then charge on each of the sphere is $\frac{a}{21} \times 10^{-8} \text{ C}$. The value of 'a' will be _____ [25 Feb, 2021 (Shift-II)]

28. A particle of mass 1 mg and charge q is lying at the mid-point of two stationary particles kept at a distance '2 m' when each is carrying same charge ' q '. If the free charged particle is displaced from its equilibrium position through distance ' x ' ($x \ll 1\text{m}$). The particle executes SHM. Its angular frequency of oscillation will be $\times 10^4 \text{ rad/s}$ if $q^2 = 10\text{C}^2$. [25 July, 2021 (Shift-I)]

29. Two identical conducting spheres with negligible volume have 2.1 nC and -0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is _____ $\times 10^{-9} \text{ N}$ [Given : $4\pi\epsilon_0 = \frac{1}{9 \times 10^9} \text{ SI Unit}$] [25 Feb, 2021 (Shift-II)]

30. An oil drop of radius 2 mm with a density 3 g cm^{-3} is held stationary under a constant electric field $3.55 \times 10^5 \text{ V m}^{-1}$ in the Millikan's oil drop experiment. What is the number of excess electrons that the oil drop will possess? (Consider $g = 9.81 \text{ m/s}^2$.) [18 March, 2021 (Shift-I)]

- (a) 1.73×10^{10} (b) 48.8×10^{11}
(c) 1.73×10^{12} (d) 17.3×10^{10}

31. An infinite number of point charges, each carrying $1 \mu\text{C}$ charge, are placed along the y-axis at, $y = 1\text{m}, 2\text{m}, 4\text{m}, 8\text{m}$. The total force on a 1 C point charge, placed at the origin, is $x \times 10^3 \text{ N}$. The value of x to the nearest integer, is _____ [Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$] [18 March, 2021 (Shift-II)]

32. Consider the force F on a charge ' q ' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F , if ' q ' is placed at distance r from the center of the shell? [6 Sep, 2020 (Shift-II)]

- (a) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$ for all r (b) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} > F > 0$ For $r < R$
(c) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$ for $r > R$ (d) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r < R$

33. Let a total charge $2Q$ be distributed in a sphere of radius R , with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B , of $-Q$ each, are placed on diametrically opposite points, at equal distance, a , from the centre. If A and B do not experience any force, then [12 April, 2019 (Shift-II)]

- (a) $a = \frac{3R}{2^{1/4}}$ (b) $a = R/\sqrt{3}$
(c) $a = 8^{-1/4} R$ (d) $a = 2^{-1/4} R$

34. Three charges $+Q, q, +Q$ are placed respectively, at distance, 0, $d/2$ and d from the origin, on the x-axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is [9 Jan, 2019 (Shift-I)]
(a) $-Q/4$ (b) $+Q/2$ (c) $+Q/4$ (d) $-Q/2$

35. Charge is distributed within a sphere of radius R with a volume charge density $\rho(r) = \frac{A}{r^2} e^{-2r/a}$, where A and a are constants. If Q is the total charge of this charge distribution the radius R is [9 Jan, 2019 (Shift-II)]

- (a) $a \log \left(1 - \frac{Q}{2\pi a A} \right)$ (b) $\frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$
(c) $a \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$ (d) $\frac{a}{2} \log \left(1 - \frac{1}{2\pi a A} \right)$

36. The bob of a simple pendulum has mass 2g and a charge of $5\mu\text{C}$. It is at rest in a uniform horizontal electric field of intensity $2 \times 10^5 \text{ V/m}$. At equilibrium, the angle that the pendulum makes with the vertical is (Take $g = 10 \text{ m/s}^2$) [8 April, 2019 (Shift-I)]

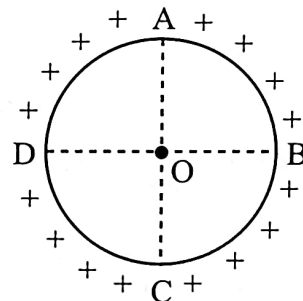
- (a) $\tan^{-1}(5.0)$ (b) $\tan^{-1}(2.0)$ (c) $\tan^{-1}(0.5)$ (d) $\tan^{-1}(0.2)$

37. The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is: [18 Mar (Shift-I)]

- (a) $\cos(\omega t) + \cos(2\omega t) + \cos(3\omega t)$
(b) $\sin^2(\omega t)$
(c) $\sin(\omega t) + \cos(\omega t)$
(d) $3 \cos \left(\frac{\pi}{4} - 2\omega t \right)$

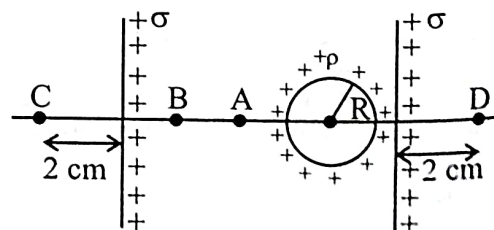
Electric Field and Field Lines

38. A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc AB to 'O' is ' E ' is magnitude. What would be the magnitude of electric field at 'O' due to arc ABC? [04 April, 2025 (Shift-I)]



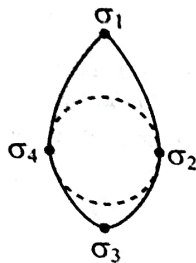
- (a) $2E$ (b) $\sqrt{2}E$
(c) $E/2$ (d) Zero

39. Two infinite identical charged sheets and a charged spherical shell of charge density ' ρ ' are arranged as shown in figure. Then the correct relation between the electrical fields at A, B, C and D points is: [04 April, 2025 (Shift-I)]



- (a) $\vec{E}_A = \vec{E}_B; \vec{E}_C = \vec{E}_D$ (b) $\vec{E}_A > \vec{E}_B; \vec{E}_C = \vec{E}_D$
(c) $\vec{E}_C \neq \vec{E}_D; \vec{E}_A > \vec{E}_B$ (d) $|\vec{E}_A| = |\vec{E}_B|; \vec{E}_C > \vec{E}_D$

40. Electric charge is transferred to an irregular metallic disk as shown in figure. If $\sigma_1, \sigma_2, \sigma_3$, and σ_4 are charge densities at given points then, choose the correct answer from the options given below:



- A. $\sigma_1 > \sigma_3; \sigma_2 = \sigma_4$ B. $\sigma_1 > \sigma_2; \sigma_3 > \sigma_4$
 C. $\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$ D. $\sigma_1 < \sigma_3 < \sigma_2 = \sigma_4$
 E. $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$

[08 April, 2025 (Shift-II)]

- (a) A, B and C Only (b) A and C Only
 (c) D and E Only (d) B and C Only

41. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The outer body of an air craft is made of metal which protects persons sitting inside from lightning-strikes.

Reason (R): The electric field inside the cavity enclosed by a conductor is zero.

In the light of the above statements, choose the most appropriate answer from the options given below: [07 April, 2025 (Shift-II)]

- (a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (b) (A) is correct but (R) is not correct
 (c) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (d) (A) is not correct but (R) is correct

42. Consider a circular loop that is uniformly charged and has a radius $a\sqrt{2}$. Find the position along the positive z-axis of the cartesian coordinate system where the electric field is maximum if the ring was assumed to be placed in xy-plane at the origin:

[02 April, 2025 (Shift-II)]

- (a) $\frac{a}{\sqrt{2}}$ (b) $\frac{a}{2}$ (c) a (d) 0

43. A point charge $+q$ is placed at the origin. A second point charge $+9q$ is placed at $(d, 0, 0)$ in Cartesian coordinate system. The point in between them where the electric field vanishes is:

[02 April, 2025 (Shift-I)]

- (a) $(4d/3, 0, 0)$ (b) $(d/4, 0, 0)$
 (c) $(3d/4, 0, 0)$ (d) $(d/3, 0, 0)$

44. Two charges q and $3q$ are separated by a distance ' r ' in air. At a distance x from charge q , the resultant electric field is zero. The value of x is:

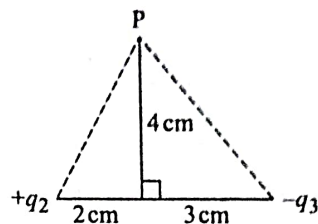
[31 Jan, 2024 (Shift-I)]

- (a) $\frac{(1+\sqrt{3})}{r}$ (b) $\frac{r}{3(1+\sqrt{3})}$ (c) $\frac{r}{(1+\sqrt{3})}$ (d) $r(1+\sqrt{3})$

45. If the net electric field at point P along Y axis is zero, then the ratio of $\frac{q_2}{q_3}$ is $\frac{8}{5\sqrt{x}}$.

where $x = \dots\dots\dots$

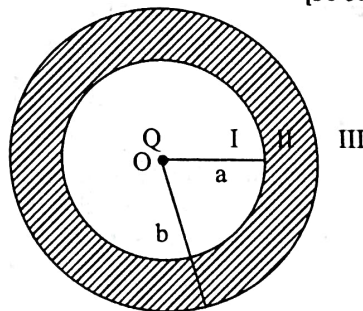
[08 April, 2024 (Shift-II)]



46. A point charge of $10\mu\text{C}$ is placed at the origin. At what location on the X-axis should a point charge of $40\mu\text{C}$ be placed so that the net electric field is zero at $x = 2\text{cm}$ on the X-axis? [25 Jan, 2023 (Shift-II)]
 (a) $x = 6\text{ cm}$ (b) $x = 4\text{ cm}$ (c) $x = 8\text{ cm}$ (d) $x = -4\text{ cm}$

47. As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b . The electric field due to charge Q in three different regions I, II and III is given by: (I: $r < a$, II: $a < r < b$, III: $r > b$)

[30 Jan, 2023 (Shift-II)]



- (a) $E_I = 0, E_{II} = 0, E_{III} \neq 0$ (b) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$
 (c) $E_I \neq 0, E_{II} = 0, E_{III} = 0$ (d) $E_I = 0, E_{II} = 0, E_{III} = 0$

48. A thin infinite sheet charge and an infinite line charge of respective charge densities $+\sigma$ and $+\lambda$ are placed parallel at 5 m distance from each other. Points 'P' and 'Q' are at $\frac{3}{\pi}$ m and $\frac{4}{\pi}$ m perpendicular distance from line charge towards sheet charge, respectively. ' E_P ' and ' E_Q ' are the magnitudes of resultant electric field intensities at point 'P' and 'Q', respectively. If $\frac{E_P}{E_Q} = \frac{4}{a}$ for $2|\sigma| = |\lambda|$. Then the value of a is _____.

[13 April, 2023 (Shift-I)]

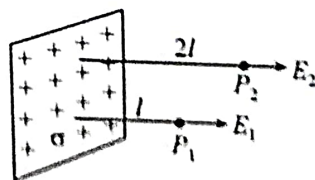
49. Two point charges A and B of magnitude $+8 \times 10^{-6}\text{ C}$ and $-8 \times 10^{-6}\text{ C}$ respectively are placed at a distance d apart. The electric field at the middle point O between the charges is $6.4 \times 10^4\text{ NC}^{-1}$. The distance ' d ' between the point charges A and B is:

[28 June, 2022 (Shift-II)]

- (a) 2.0 m (b) 3.0 m (c) 1.0 m (d) 4.0 m

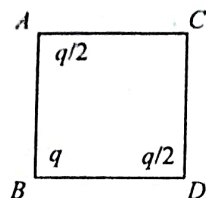
50. In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance l and $2l$ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are

[25 June, 2022 (Shift-I)]



- (a) $E_1 = \sigma/\epsilon_0, E_2 = \sigma/2\epsilon_0$ (b) $E_1 = 2\sigma/\epsilon_0, E_2 = \sigma/\epsilon_0$
 (c) $E_1 = E_2 = \sigma/2\epsilon_0$ (d) $E_1 = E_2 = \sigma/\epsilon_0$

51. The three charges $q/2$, q and $q/2$ are placed at the corners A , B and C of a square of side ' a ' as shown in figure. The magnitude of electric field (E) at the corner D of the square, is [28 June, 2022 (Shift-I)]



- (a) $\frac{q}{4\pi\epsilon_0 a^2} \left(\frac{1}{\sqrt{2}} + \frac{1}{2} \right)$ (b) $\frac{q}{4\pi\epsilon_0 a^2} \left(1 + \frac{1}{\sqrt{2}} \right)$
 (c) $\frac{q}{4\pi\epsilon_0 a^2} \left(1 - \frac{1}{\sqrt{2}} \right)$ (d) $\frac{q}{4\pi\epsilon_0 a^2} \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right)$

52. An oil drop of radius 2 mm with a density 3 g cm^{-3} is held stationary under a constant electric field $3.55 \times 10^5 \text{ V m}^{-1}$ in the Millikan's oil drop experiment. What is the number of excess electrons that the oil drop will possess? (Consider $g = 9.81 \text{ m/s}^2$)

[18 March, 2021 (Shift-I)]

- (a) 1.73×10^{10} (b) 48.8×10^{11}
 (c) 1.73×10^{12} (d) 17.3×10^{10}

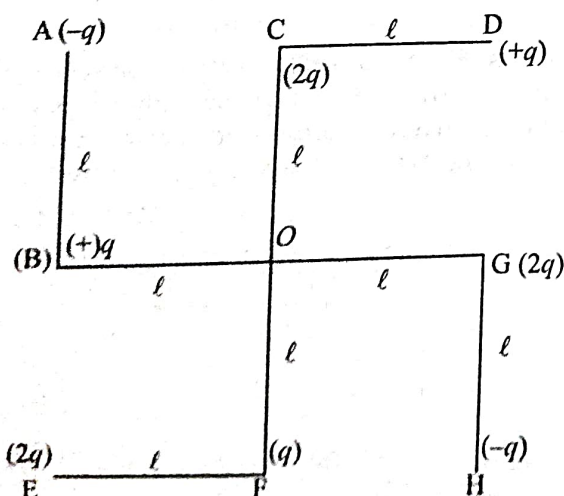
53. A uniformly charged disc of radius R having surface charge density σ is placed in the xy plane with its center at the origin. Find the electric field intensity along the z -axis at a distance Z from origin:

[27 Aug, 2021 (Shift-I)]

- (a) $E = \frac{\sigma}{2\epsilon_0} \left(1 + \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$
 (b) $E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$
 (c) $E = \frac{2\epsilon_0}{\sigma} \left(\frac{1}{(Z^2 + R^2)^{1/2}} + Z \right)$
 (d) $E = \frac{\sigma}{2\epsilon_0} \left(\frac{1}{(Z^2 + R^2)^{1/2}} + \frac{1}{Z^2} \right)$

54. What will be the magnitude of electric field at point O as shown in figure? Each side of the figure is ℓ and perpendicular to each other?

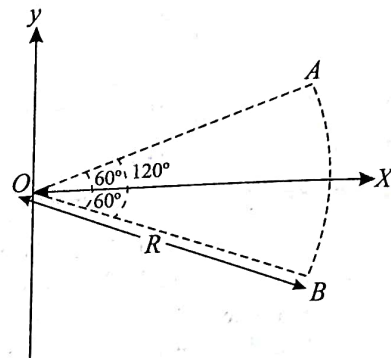
[27 July, 2021 (Shift-II)]



- (a) $\frac{1}{4\pi\epsilon_0} \frac{q}{(2\ell^2)} (2\sqrt{2} - 1)$ (b) $\frac{q}{4\pi\epsilon_0 (2\ell)^2}$
 (c) $\frac{1}{4\pi\epsilon_0} \frac{q}{\ell^2}$ (d) $\frac{1}{4\pi\epsilon_0} \frac{2q}{2\ell^2} (\sqrt{2})$

55. Figure shows a rod AB , which is bent in a 120° circular arc of radius R . A charge $(-Q)$ is uniformly distributed over rod AB . What is the electric field E at the centre of curvature O ?

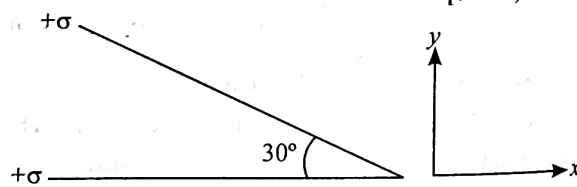
[27 Aug, 2021 (Shift-II)]



- (a) $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2} (-\hat{i})$ (b) $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2} (\hat{i})$
 (c) $\frac{3\sqrt{3}Q}{8\pi\epsilon_0 R^2} (\hat{i})$ (d) $\frac{3\sqrt{3}Q}{16\pi^2\epsilon_0 R^2} (\hat{i})$

56. Two infinite planes each with uniform surface charge density $+\sigma$ are kept in such a way that the angle between them is 30° . The electric field in the region shown between them is given by:

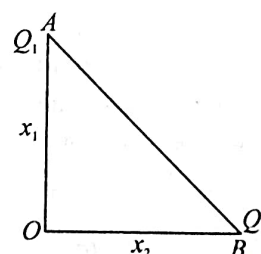
[7 Jan, 2020 (Shift-I)]



- (a) $\frac{\sigma}{2\epsilon_0} \left[\left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} - \frac{\hat{x}}{2} \right]$ (b) $\frac{\sigma}{2\epsilon_0} \left[\left(1 + \sqrt{3} \right) \hat{y} - \frac{\hat{x}}{2} \right]$
 (c) $\frac{\sigma}{2\epsilon_0} \left[\left(1 + \sqrt{3} \right) \hat{y} + \frac{\hat{x}}{2} \right]$ (d) $\frac{\sigma}{\epsilon_0} \left[\left(1 + \frac{\sqrt{3}}{2} \right) \hat{y} + \frac{\hat{x}}{2} \right]$

57. Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1/Q_2 is proportional to

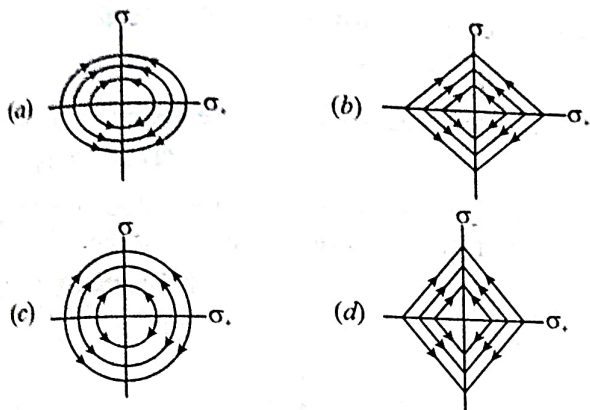
[6 Sep, 2020 (Shift-I)]



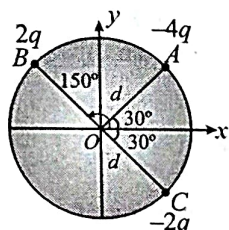
- (a) $\frac{x_1^3}{x_2^3}$ (b) $\frac{x_2^2}{x_1^2}$ (c) $\frac{x_1}{x_2}$ (d) $\frac{x_2}{x_1}$

58. Two charged thin infinite plane sheets of uniform surface charge density σ_1 and σ_2 , where $|\sigma_1| > |\sigma_2|$, intersect at right angle. Which of the following best represents the electric field lines for this system?

[4 Sep, 2020 (Shift-I)]

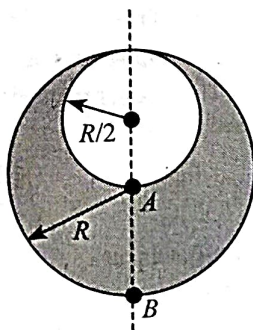


59. Three charged particles A , B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d . The charged particles A , C and center O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x -direction is [8 Jan, 2020 (Shift-I)]



- (a) $\frac{\sqrt{3}q}{\pi\epsilon_0 d^2}$ (b) $\frac{\sqrt{3}q}{4\pi\epsilon_0 d^2}$ (c) $\frac{3\sqrt{3}q}{4\pi\epsilon_0 d^2}$ (d) $\frac{2\sqrt{3}q}{\pi\epsilon_0 d^2}$

60. Consider a sphere of radius R which carries a uniform charge density ρ . If a sphere of radius $R/2$ is carved out of it, as shown the ratio $\frac{|\vec{E}_A|}{|\vec{E}_B|}$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at point A and B due to the remaining portion is: [9 Jan, 2020 (Shift-I)]



- (a) $\frac{18}{34}$ (b) $\frac{17}{54}$ (c) $\frac{18}{54}$ (d) $\frac{21}{34}$

61. Two point charges $q_1 (\sqrt{10} \mu C)$ and $q_2 (-25 \mu C)$ are placed on the x -axis at $x = 1$ m and $x = 4$ m respectively. The electric field (in V/m) at a point $y = 3$ m on y -axis is, [9 Jan, 2019 (Shift-II)]

[take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$]

- (a) $(63\hat{i} - 27\hat{j}) \times 10^2$ (b) $(-63\hat{i} - 27\hat{j}) \times 10^2$
(c) $(81\hat{i} - 81\hat{j}) \times 10^2$ (d) $(-81\hat{i} - 81\hat{j}) \times 10^2$

62. For a uniformly charged ring of radius R , the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is [9 Jan, 2019 (Shift-I)]

- (a) $\frac{R}{\sqrt{5}}$ (b) $\frac{R}{\sqrt{2}}$ (c) R (d) $R\sqrt{2}$

63. Four point charges $-q$, $+q$, $+q$ and $-q$ are placed on y -axis at $y = -2d$, $y = -d$, $y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x -axis at $x = D$, with $D \gg d$, will vary as [9 April, 2019 (Shift-II)]

- (a) $E \propto \frac{1}{D}$ (b) $E \propto \frac{1}{D^3}$ (c) $E \propto \frac{1}{D^2}$ (d) $E \propto \frac{1}{D^4}$

Electric Dipole, Force & Torque on Dipole

64. Two small spherical balls of mass 10 g each with charges $-2\mu C$ and $2\mu C$, are attached to two ends of very light rigid rod of length 20 cm. The arrangement is now placed near an infinite nonconducting charge sheet with uniform charge density of $100\mu C/m^2$ such that length of rod makes an angle of 30° with electric field generated by charge sheet. Net torque acting on the rod is:

(Take $\epsilon_0 : 8.85 \times 10^{-12} \text{ C}^2 / \text{Nm}^2$) [04 April, 2025 (Shift-I)]

- (a) 112 Nm (b) 1.12 Nm (c) 2.24 Nm (d) 11.2 Nm

65. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Net dipole moment of a polar linear isotropic dielectric substance is not zero even in the absence of an external electric field.

Reason (R): In absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions.

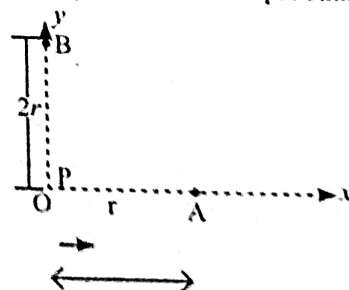
In the light of the above statements, choose the most appropriate answer from the options given below: [02 April, 2025 (Shift-II)]

- (a) (A) is correct but (R) is not correct
(b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(c) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(d) (A) is not correct but (R) is correct

66. A dipole with two electric charges of $2\mu C$ magnitude each, with separation distance 0.5 μm , is placed between the plates of a capacitor such that its axis is parallel to an electric field established between the plates when a potential difference of 5 V is applied. Separation between the plates is 0.5 mm. If the dipole is rotated by 30° from the axis, it tends to realign in the direction due to a torque. The value of torque is: [07 April, 2025 (Shift-II)]

- (a) $5 \times 10^{-9} \text{ Nm}$ (b) $5 \times 10^{-3} \text{ Nm}$
(c) $2.5 \times 10^{-12} \text{ Nm}$ (d) $2.5 \times 10^{-9} \text{ Nm}$

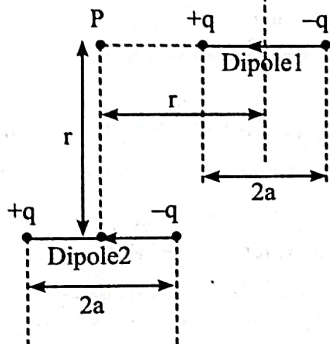
67. For a short dipole placed at origin O , the dipole moment P is along x -axis, as shown in the figure. If the electric potential and electric field at A are V_0 and E_0 , respectively, then the correct combination of the electric potential and electric field, respectively, at point B on the y -axis is given by: [22 Jan, 2025 (Shift-II)]



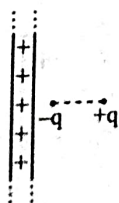
- (a) zero and $\frac{E_0}{16}$ (b) $\frac{V_0}{2}$ and $\frac{E_0}{16}$
 (c) zero and $\frac{E_0}{8}$ (d) V_0 and $\frac{E_0}{4}$

68. A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a distance r . The dipoles are made of opposite charge q separated by a distance $2a$. For the charge particle at P not to experience any net force, which of the following correctly describes the situation?

[23 Jan, 2025 (Shift-I)]



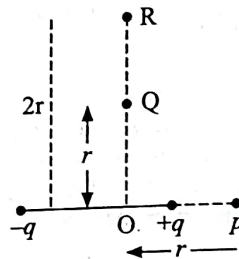
- (a) $\frac{a}{r} \sim 3$ (b) $\frac{a}{r} \sim 20$ (c) $\frac{a}{r} \sim 0.5$ (d) $\frac{a}{r} \sim 10$
69. Two point charges $-4 \mu\text{C}$ and $4 \mu\text{C}$, constituting an electric dipole, are placed at $(-9, 0, 0)$ cm and $(9, 0, 0)$ cm in a uniform electric field of strength 10^4 NC^{-1} . The work done on the dipole in rotating it from the equilibrium through 180° is: [23 Jan, 2025 (Shift-II)]
 (a) 16.4 mJ (b) 12.4 mJ (c) 18.4 mJ (d) 14.4 mJ
70. An electric dipole of dipole moment $6 \times 10^{-6} \text{ Cm}$ is placed in uniform electric field of magnitude 10^6 V/m . Initially, the dipole moment is parallel to electric field. The work that needs to be done on the dipole to make its dipole moment opposite to the field, will be _____ J. [28 Jan, 2025 (Shift-II)]
71. An electric dipole of mass m , charge q , and length l is placed in a uniform electric field $\vec{E} = E_0 \hat{i}$. When the dipole is rotated slightly from its equilibrium position and released, the time period of its oscillations will be: [29 Jan, 2025 (Shift-I)]
 (a) $\frac{1}{2\pi} \sqrt{\frac{2ml}{qE_0}}$ (b) $2\pi \sqrt{\frac{ml}{qE_0}}$ (c) $2\pi \sqrt{\frac{ml}{2qE_0}}$ (d) $\frac{1}{2\pi} \sqrt{\frac{ml}{2qE_0}}$
72. An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0 . Choose the correct option from the following. [29 Jan, 2025 (Shift-II)]



- (a) Potential energy and torque both are maximum.
 (b) Torque on dipole is zero and net force acts towards the sheet.
 (c) Potential energy of dipole is minimum and torque is zero.
 (d) Torque on dipole is zero and net force is directed away from the sheet.

73. Two charges of $-4 \mu\text{C}$ and $+4 \mu\text{C}$ are placed at the points $A(1, 0, 4)$ m and $B(2, -1, 5)$ located in an electric field $\vec{E} = 0.20 \hat{i} \text{ V/cm}$. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5} \text{ Nm}$ where $\alpha =$ _____. [27 Jan, 2024 (Shift-II)]

74. The electric field at point p due to an electric dipole is E . The electric field at point R on equatorial line will be E/x . The value of x is: [05 April, 2024 (Shift-II)]



75. The electric field due to a short electric dipole at a large distance r from center of dipole on the equatorial plane varies with distance as: [15 April, 2023 (Shift-I)]

- (a) r (b) $\frac{1}{r}$ (c) $\frac{1}{r^3}$ (d) $\frac{1}{r^2}$

76. Two charges each of magnitude 0.01 C and separated by a distance of 0.4 mm constitute an electric dipole. If the dipole is placed in a uniform electric field ' $\vec{E}$ ' of 10 dyne/C making 30° angle with $\vec{E}$, the magnitude of torque acting on dipole is [13 April, 2023 (Shift-I)]

- (a) $4.0 \times 10^{-10} \text{ Nm}$ (b) $2.0 \times 10^{-10} \text{ Nm}$
 (c) $1.0 \times 10^{-8} \text{ Nm}$ (d) $1.5 \times 10^{-9} \text{ Nm}$

77. A dipole comprises of two charged particles of identical magnitude and opposite in nature. The mass m of the positive charged particle is half of the mass of the negative charged particle. The two charges are separated by a distance l . If the dipole is placed in a uniform electric field E ; such a way that dipole axis makes a very small angle with the electric field E . The angular frequency of the oscillations of the dipole when released is given by: [6 April, 2023 (Shift-II)]

- (a) $\sqrt{\frac{8qqE}{3ml}}$ (b) $\sqrt{\frac{4qE}{ml}}$ (c) $\sqrt{\frac{4qE}{3ml}}$ (d) $\sqrt{\frac{8qE}{ml}}$

78. Given below are two statements:

Statement-I: A point charge is brought in an electric field. The value of electric field at a point near to the charge may increase if the charge is positive.

Statement-II: An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero. Choose the correct answer from the options given below:

[28 June, 2022 (Shift-I)]

- (a) Both Statement-I and Statement-II are true
 (b) Both Statement-I and Statement-II are false
 (c) Statement-I is true but Statement-II is false
 (d) Statement-I is false but Statement-II is True

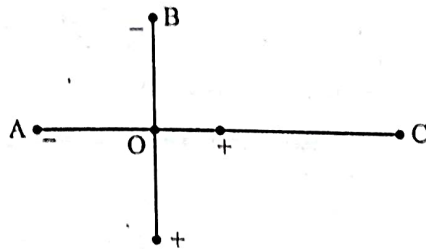
79. Two electric dipoles of dipole moments 1.2×10^{-30} and $2.4 \times 10^{-30} \text{ Cm}$ are placed in two different uniform electric fields of strengths $5 \times 10^4 \text{ NC}^{-1}$ and $15 \times 10^4 \text{ NC}^{-1}$ respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$. The value of x is _____. [28 July, 2022 (Shift-I)]

80. Two ideal electric dipoles A and B , having their dipole moment p_1 and p_2 respectively are placed on a plane with their centres at O as shown in the figure. At point C on the axis of dipole A , the resultant electric field is making an angle of 37° with the axis.

The ratio of the dipole moment of A and B , $\frac{p_1}{p_2}$ is:

(take $\sin 37^\circ = \frac{3}{5}$)

[25 July, 2021 (Shift-II)]



- (a) $\frac{3}{8}$ (b) $\frac{3}{2}$ (c) $\frac{4}{3}$ (d) $\frac{2}{3}$

81. An electric dipole is placed on x -axis in proximity to a line charge of linear density $3.0 \times 10^{-6} \text{ C/m}$. Line charge is placed on z -axis and positive and negative charge of dipole is at a distance of 10 mm and 12 mm from the origin respectively. If total force of 4 N is exerted on the dipole, find out the amount of positive charge of the dipole.

[22 July, 2021 (Shift-II)]

- (a) 0.485 mC (b) 8.8 μC (c) 815.1 nC (d) 4.44 μC

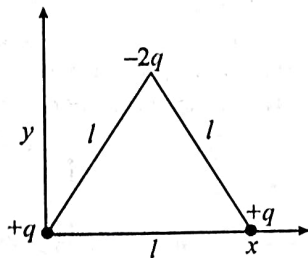
82. An electric dipole of moment $\vec{p} = (-\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29} \text{ C.m.}$ is at the origin (0, 0, 0). The electric field due to this dipole at $\vec{r} = (+\hat{i} + 3\hat{j} + 5\hat{k})$ (note that $\vec{r} \cdot \vec{p} = 0$) is parallel to

[9 Jan, 2020 (Shift-I)]

- (a) $(+\hat{i} + 3\hat{j} - 2\hat{k})$ (b) $(-\hat{i} + 3\hat{j} - 2\hat{k})$
(c) $(+\hat{i} - 3\hat{j} - 2\hat{k})$ (d) $(-\hat{i} - 3\hat{j} + 2\hat{k})$

83. Determine the electric dipole moment of the system of three charges, placed on the vertices of an equilateral triangle, as shown in the figure

[12 Jan, 2019 (Shift-I)]



- (a) $\sqrt{3}ql \frac{\hat{j} - \hat{i}}{\sqrt{2}}$ (b) $(ql) \frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (c) $2ql\hat{j}$ (d) $-\sqrt{3}ql\hat{j}$

84. An electric dipole is formed by two equal and opposite charges q with separation d . The charges have same mass m . It is kept in a uniform electric field E . If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is

[8 April, 2019 (Shift-II)]

- (a) $\sqrt{\frac{qE}{2md}}$ (b) $2\sqrt{\frac{qE}{md}}$ (c) $\sqrt{\frac{2qE}{md}}$ (d) $\sqrt{\frac{qE}{md}}$

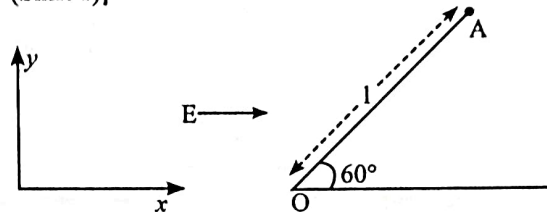
Motion of Charged Particle in Uniform Electric Field

85. An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerges out of the electric field region with a horizontal component of velocity 10^6 m/s . If the magnitude of

the electric field between the plates is 9.1 V/cm , then the vertical component of velocity of electron is (mass of electron $= 9.1 \times 10^{-31} \text{ kg}$ and charge of electron $= 1.6 \times 10^{-19} \text{ C}$) [22 Jan, 2025 (Shift-I)]

- (a) $16 \times 10^6 \text{ m/s}$ (b) 0
(c) $16 \times 10^4 \text{ m/s}$ (d) $1 \times 10^6 \text{ m/s}$

86. A particle of mass ' m ' and charge ' q ' is fastened to one end ' A ' of a massless string having equilibrium length ' l ', whose other end is fixed at point ' O '. The whole system is placed on a frictionless horizontal plane and is initially at rest. If uniform electric field is switched on along the direction as shown in figure, then the speed of the particle when it crosses the x -axis is [28 Jan, 2025 (Shift-I)]



- (a) $\sqrt{\frac{2qEl}{m}}$ (b) $\sqrt{\frac{qEl}{m}}$ (c) $\sqrt{\frac{qEl}{4m}}$ (d) $\sqrt{\frac{qEl}{2m}}$

87. A particle of charge ' $-q$ ' and mass ' m ' moves in a circle of radius ' r ' around an infinitely long line charge of linear density ' $+\lambda$ '. Then time period will be given as: (Consider k as Coulomb's constant)

[30 Jan, 2024 (Shift-II)]

- (a) $T^2 = \frac{4\pi m}{2k\lambda q} r^3$ (b) $T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$
(c) $T = \frac{1}{2\pi r} \sqrt{\frac{m}{2k\lambda q}}$ (d) $T = \frac{1}{2\pi} \sqrt{\frac{2k\lambda q}{m}}$

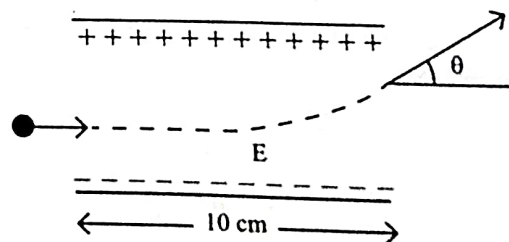
88. An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density $+\sigma$. The electron at $t = 0$ is at a distance of 1 m from S and has a speed of 1 m/s. The maximum value of σ if the electron strikes

S at $t = 1 \text{ s}$ is $\propto \left[\frac{m\epsilon_0}{e} \right] \frac{C}{m^2}$ the value of α is

[29 Jan, 2024 (Shift-I)]

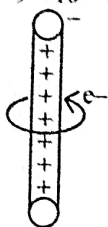
89. A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV . The length of each plate is 10 cm . The angle (θ) of deviation of the path of electron as it comes out of the field is (in degree).

[25 Jan, 2023 (Shift-I)]



90. A stream of a positively charged particles having $\frac{q}{m} = 2 \times 10^{11} \frac{\text{C}}{\text{kg}}$ and velocity $\vec{v}_0 = 3 \times 10^7 \hat{i} \text{ m/s}$ is deflected by an electric field $1.8 \hat{j} \text{ kV/m}$. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charge particles in the y direction is _____ mm. [24 Jan, 2023 (Shift-I)]

91. An electron revolves around an infinite cylindrical wire having uniform linear charge density $2 \times 10^{-8} \text{ cm}^{-1}$ in circular path under the influence of attractive electrostatic field as shown in the figure. The velocity of electron with which it is revolving is $\times 10^6 \text{ ms}^{-1}$. Given mass of electron $= 9 \times 10^{-31} \text{ kg}$. [10 April, 2023 (Shift-II)]



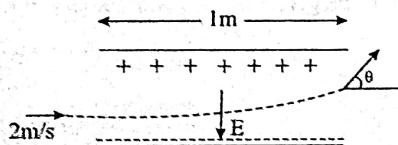
92. A vertical electric field of magnitude $4.9 \times 10^5 \text{ N/C}$ just prevents a water droplet of a mass 0.1 g from falling. The value of charge on the droplet will be:

(Given $g = 9.8 \text{ m/s}^2$)

[24 June, 2022 (Shift-I)]

- (a) $1.6 \times 10^{-9} \text{ C}$ (b) $2.0 \times 10^{-9} \text{ C}$
(c) $3.2 \times 10^{-9} \text{ C}$ (d) $0.5 \times 10^{-9} \text{ C}$

93. A uniform electric field $E = (8m/e) \text{ V/m}$ is created between two parallel plates of length 1 m as shown in figure, (where $m =$ mass of electron and $e =$ charge of electron). An electron enters the field symmetrically between the plates with a speed of 2 m/s . The angle of the deviation (θ) of the path of the electron as it comes out of the field will be _____. [28 July, 2022 (Shift-II)]



- (a) $\tan^{-1}(4)$ (b) $\tan^{-1}(2)$ (c) $\tan^{-1}\left(\frac{1}{3}\right)$ (d) $\tan^{-1}(3)$

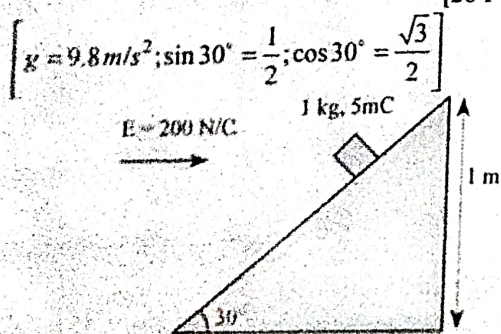
94. A long cylindrical volume contains a uniformly distributed charge of density ρ . The radius of cylindrical volume is R . A charge particle (q) revolves around the cylinder in a circular path. The kinetic energy of the particle is: [24 June, 2022 (Shift-II)]

- (a) $\frac{\rho q R^2}{4\epsilon_0}$ (b) $\frac{\rho q R^2}{2\epsilon_0}$ (c) $\frac{q \rho}{4\epsilon_0 R^2}$ (d) $\frac{4\epsilon_0 R^2}{q \rho}$

95. A positive charge particle of 100 mg is thrown in opposite direction to a uniform electric field of strength $1 \times 10^5 \text{ NC}^{-1}$. If the charge on the particle is $40 \mu\text{C}$ and the initial velocity is 200 ms^{-1} , how much distance it will travel before coming to the rest momentarily: [29 June, 2022 (Shift-I)]

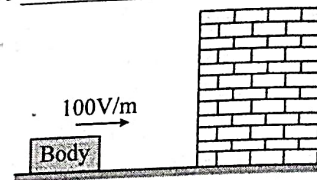
- (a) 1 m (b) 5 m (c) 10 m (d) 0.5 m

96. An inclined plane making an angle of 30° with the horizontal is placed in a uniform horizontal electric field $200 \frac{\text{N}}{\text{C}}$ as shown in the figure. A body of mass 1 kg and charge 5 mC is allowed to slide down from rest at a height of 1 m . If the coefficient of friction is 0.2 , find the time taken by the body to reach the bottom. [26 Feb, 2021 (Shift-II)]

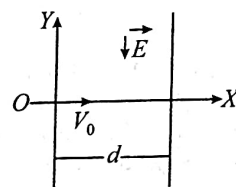


- (a) 1.3 s (b) 0.46 s (c) 2.3 s (d) 0.92 s

97. A body having specific charge $8 \mu\text{C/g}$ is resting on a frictionless plane at a distance 10 cm from the wall (as shown in the figure). It starts moving towards the wall when a uniform electric field of 100 V/m is applied horizontally towards the wall. If the collision of the body with the wall is perfectly elastic, then the time period of its motion will be _____. [20 July, 2021 (Shift-I)]



98. A charged particle (mass m and charge q) moves along X -axis with velocity V_0 . When it passes through the origin it enters a region having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is [2 Sep, 2020 (Shift-I)]

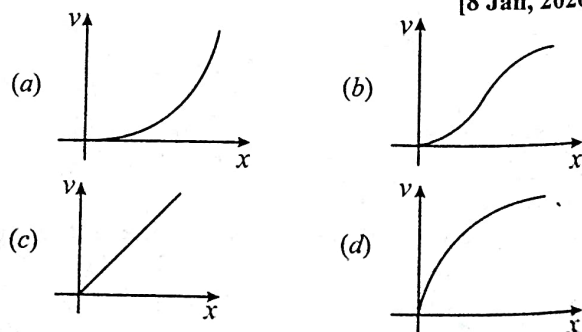


- (a) $y = \frac{qEd}{mV_0^2}(x-d)$ (b) $y = \frac{qEd^2}{mV_0^2}x$
(c) $y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$ (d) $y = \frac{qEd}{mV_0^2}x$

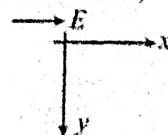
99. A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x -direction, where a and E_0 are constants. Initially the particle was at rest at $x = 0$. Other than the initial position, the kinetic energy of the particle becomes zero when the distance of the particle from the origin is [4 Sep, 2020 (Shift-II)]

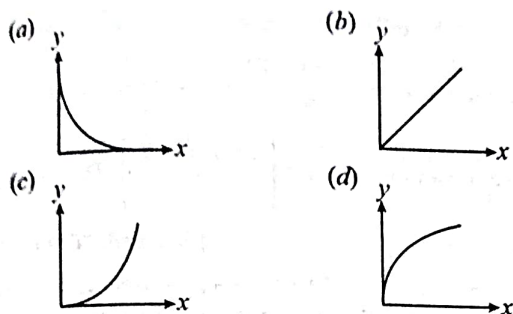
- (a) $\sqrt{\frac{3}{a}}$ (b) $\sqrt{\frac{2}{a}}$ (c) $\sqrt{\frac{1}{a}}$ (d) a

100. A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale) [8 Jan, 2020 (Shift-II)]



101. A small point mass carrying some positive charge on it, is released from the edge of a table. There is a uniform electric field in this region in the horizontal direction. Which of the following options then correctly describe the trajectory of the mass? (Curves are drawn schematically and are not to scale). [2 Sep, 2020 (Shift-I)]





Gauss Law and Application, Electric Flux

102. An infinitely long wire has uniform linear charge density $\lambda = 2\text{ nC/m}$. The net flux through a Gaussian cube of side length $\sqrt{3}\text{ cm}$, if the wire passes through any two corners of the cube, that are maximally displaced from each other, would be $x\text{ Nm}^2\text{ C}^{-1}$, where x is:

[Neglect any edge effects and use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\text{ SI units}$]

[08 April, 2025 (Shift-II)]

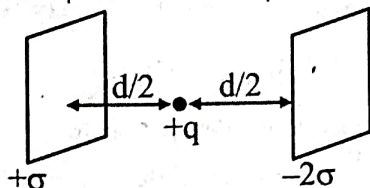
- (a) 0.72π (b) 1.44π (c) 6.48π (d) 2.16π

103. The electric field in a region is given by $\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3\text{ N/C}$.

The flux of the field through a rectangular surface parallel to $x-z$ plane is $6.0\text{ Nm}^2\text{ C}^{-1}$. The area of the surface is cm^2 .

[07 April, 2025 (Shift-II)]

104. Consider two infinitely large plane parallel conducting plates as shown below. The plates are uniformly charged with a surface charge density $+\sigma$ and -2σ . The force experienced by a point charge $+q$ placed at the mid point between two plates will be:

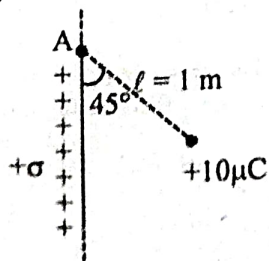


[02 April, 2025 (Shift-I)]

- (a) $\frac{\sigma q}{4\epsilon_0}$ (b) $\frac{3\sigma q}{2\epsilon_0}$ (c) $\frac{3\sigma q}{4\epsilon_0}$ (d) $\frac{\sigma q}{2\epsilon_0}$

105. A small bob of mass 100 mg and charge $+10\mu\text{C}$ is connected to an insulating string of length 1 m . It is brought near to an infinitely long non-conducting sheet of charge density $+\sigma$ as shown in figure. If string subtends an angle of 45° with the sheet at equilibrium the charge density of sheet will be:

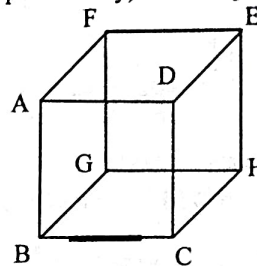
(Given, $\epsilon_0 = 8.85 \times 10^{-12}\text{ F/m}$ and acceleration due to gravity, $g = 10\text{ m/s}^2$)



[02 April, 2025 (Shift-I)]

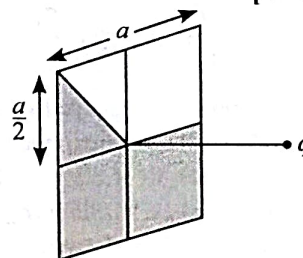
- (a) 0.885 nC/m^2 (b) 17.7 nC/m^2
(c) 885 nC/m^2 (d) 1.77 nC/m^2

106. A line charge of length $\frac{a'}{2}$ is kept at the center of an edge BC of a cube ABCDEFGH having edge length a' as shown in the figure. If the density of line charge is $\lambda\text{ C per unit length}$, then the total electric flux through all the faces of the cube will be _____. (Take, ϵ_0 as the free space permittivity). [22 Jan, 2025 (Shift-I)]



- (a) $\frac{\lambda a}{16\epsilon_0}$ (b) $\frac{\lambda a}{2\epsilon_0}$ (c) $\frac{\lambda a}{4\epsilon_0}$ (d) $\frac{\lambda a}{8\epsilon_0}$

107. A square loop of sides $a = 1\text{ m}$ is held normally in front of a point charge $q = 1\text{ C}$. The flux of the electric field through the shaded region is $\frac{5}{p} \times \frac{1}{\epsilon_0} \frac{\text{Nm}^2}{\text{C}}$, where the value of p is _____. [24 Jan, 2025 (Shift-I)]



108. Match List-I with List-II.

[29 Jan, 2025 (Shift-I)]

List-I		List-II	
A.	Electric field inside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ , and radius R .	I.	$\frac{\sigma}{\epsilon_0}$
B.	Electric field at distance $r > 0$ from a uniformly charged infinite plane sheet with surface charge density σ .	II.	$\frac{\sigma}{2\epsilon_0}$
C.	Electric field outside (distance $r > 0$ from center) of a uniformly charged spherical shell with surface charge density σ , and radius R .	III.	0
D.	Electric field between 2 oppositely charged infinite plane parallel sheets with uniform surface charge density σ .	IV.	$\frac{\sigma}{\epsilon_0 r^2}$

Choose the correct answer from the options given below:

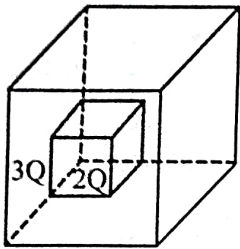
- (a) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
(b) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(c) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)
(d) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

109. A point charge causes an electric flux of $-2 \times 10^4\text{ Nm}^2\text{ C}^{-1}$ to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is:

(Given $\epsilon_0 = 8.85 \times 10^{-12}\text{ C}^2\text{ N}^{-1}\text{ m}^{-2}$) [29 Jan, 2025 (Shift-II)]

- (a) $17.7 \times 10^{-8}\text{ C}$ (b) $-15.7 \times 10^{-8}\text{ C}$
(c) $-17.7 \times 10^{-8}\text{ C}$ (d) $15.7 \times 10^{-8}\text{ C}$

110. C_1 and C_2 are two hollow concentric cubes enclosing charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is:



[1 Feb, 2024 (Shift-II)]

- (a) 2 : 5 (b) 5 : 2 (c) 2 : 3 (d) 3 : 2
111. An electric field is given by $(6\hat{i} + 5\hat{j} + 3\hat{k})$ N/C. The electric flux through a surface area $30\hat{i}$ m² lying in YZ-plane (in SI unit) is:

[29 Jan, 2024 (Shift-II)]

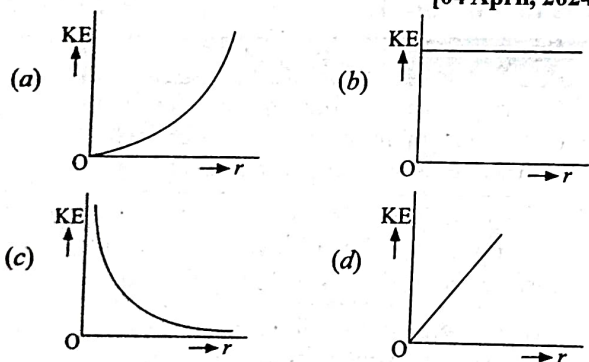
- (a) 90 (b) 150 (c) 180 (d) 60
112. Two charges of $5Q$ and $-2Q$ are situated at the points $(3a, 0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius $'4a'$ having center at origin is:

[29 Jan, 2024 (Shift-I)]

- (a) $\frac{2Q}{\epsilon_0}$ (b) $\frac{5Q}{\epsilon_0}$ (c) $\frac{7Q}{\epsilon_0}$ (d) $\frac{3Q}{\epsilon_0}$

113. An infinitely long positively charged straight thread has a linear charge density λ C m⁻¹. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is:

[04 April, 2024 (Shift-I)]



114. An infinite plane sheet of charge having uniform surface charge density $+\sigma$ C/m² is placed on x-y plane. Another infinitely long line charge having uniform linear charge density $+\lambda$ C/m is placed at $z = 4$ m plane and parallel to y-axis. If the magnitude values $|\sigma| = 2|\lambda|$, then at point $(0, 0, 2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$.

The value of n is _____.

[04 April, 2024 (Shift-I)]

115. A charge q is placed at the center of one of the surface of a cube. The flux linked with the cube is:

[04 April, 2024 (Shift-II)]

- (a) $\frac{q}{4\epsilon_0}$ (b) $\frac{q}{2\epsilon_0}$ (c) $\frac{q}{8\epsilon_0}$ (d) Zero

116. σ is the uniform surface charge density of a thin spherical shell of radius R . The electric field at any point on the surface of the spherical shell is:

[06 April, 2024 (Shift-I)]

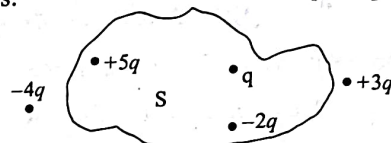
- (a) $\sigma/\epsilon_0 R$ (b) $\sigma/2\epsilon_0$ (c) σ/ϵ_0 (d) $\sigma/4\epsilon_0$

117. An electric field, $\vec{E} = \frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}}$ passes through the surface of 4 m² area having unit vector $\hat{n} = \left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ V m.

[08 April, 2024 (Shift-I)]

118. Five charges $+q, +5q, -2q, +3q$ and $-4q$ are situated as shown in the figure. The electric flux due to this configuration through the surface S is:

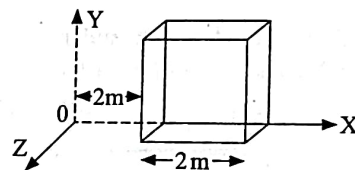
[09 April, 2024 (Shift-II)]



- (a) $\frac{5q}{\epsilon_0}$ (b) $\frac{4q}{\epsilon_0}$ (c) $\frac{3q}{\epsilon_0}$ (d) $\frac{q}{\epsilon_0}$

119. An electric field $\vec{E} = 2x\hat{i}$ NC⁻¹ exists in space. A cube of side 2 m is placed in the space as per figure given below. The electric flux through the cube is _____ Nm²/C.

[09 April, 2024 (Shift-II)]



120. In a cuboid of dimension $2L \times 2L \times L$, a charge q is placed at the centre of the surface 'S' having area of $4L^2$. The flux through the opposite surface to 'S' is given by

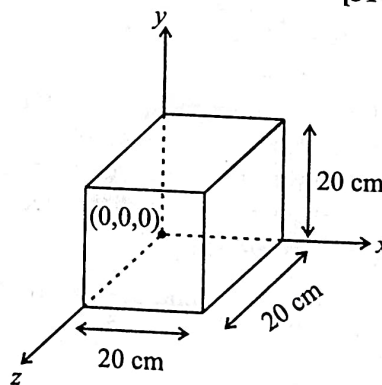
[29 Jan, 2023 (Shift-I)]

- (a) $\frac{q}{12\epsilon_0}$ (b) $\frac{q}{3\epsilon_0}$ (c) $\frac{q}{2\epsilon_0}$ (d) $\frac{q}{6\epsilon_0}$

121. Expression for an electric field is given by

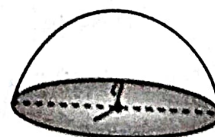
$\vec{E} = 4000x^2\hat{i}$ V/m. The electric flux through the cube of side 20 cm when placed in electric field (as shown in the figure) is _____ V cm.

[31 Jan, 2023 (Shift-I)]



122. If a charge q is placed at the center of a closed hemi-spherical non-conducting surface, the total flux passing through the flat surface would be

[27 June, 2022 (Shift-II)]

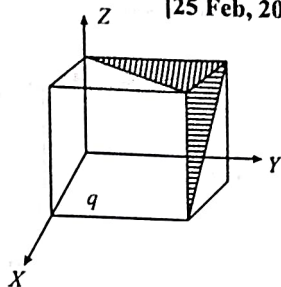


- (a) $\frac{q}{\epsilon_0}$ (b) $\frac{q}{2\epsilon_0}$ (c) $\frac{q}{4\epsilon_0}$ (d) $\frac{q}{2\pi\epsilon_0}$

123. A charge 'q' is placed at one corner of a cube as shown in figure. The flux of electrostatic field E through the shaded area is:

[25 Feb, 2021 (Shift-II)]

- (a) $\frac{q}{48\epsilon_0}$
 (b) $\frac{q}{8\epsilon_0}$
 (c) $\frac{q}{24\epsilon_0}$
 (d) $\frac{q}{4\epsilon_0}$



124. Given below are two statements:

Statement-I: An electric dipole is placed at the center of a hollow sphere. The flux of electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.

Statement-II: If R is the radius of a solid metallic sphere and Q be the total charge on it. The electric field at any point on the spherical surface of radius r ($r < R$) is zero but the electric flux passing through this closed spherical surface of radius r is not zero.

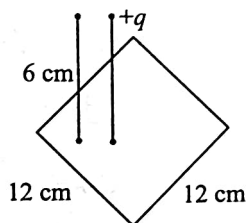
In the light of the above statements. Choose the correct answer from the option given below:

[26 Feb, 2021 (Shift-II)]

- (a) Statement-I is true but Statement-II is false
 (b) Statement-I is false but Statement-II is true
 (c) Both Statement-I and Statement-II are true
 (d) Both Statement-I and Statement-II are false

125. A point charge of $+12\mu\text{C}$ is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be $\times 10^3 \text{ Nm}^2/\text{C}$.

[24 Feb, 2021 (Shift-II)]



126. The total charge enclosed in an incremental volume of $2 \times 10^{-9} \text{ m}^3$ located at the origin is nC , if electric flux density of its field is found as

[22 July, 2021 (Shift-II)]

$$\vec{D} = e^{-x} \sin y \hat{i} - e^{-x} \cos y \hat{j} + 2z \hat{k} \text{ C/m}^2$$

127. The electric field in a region is given by

$$\vec{E} = \left(\frac{3}{5} E_0 \hat{i} + \frac{4}{5} E_0 \hat{j} \right) \frac{N}{C}$$

The ratio of flux of reported field through the rectangular surface of area 0.2 m^2 (parallel to $y-z$ plane) to that of the surface of area 0.3 m^2 (parallel to $x-z$ plane) is $a:b$, where $a =$ (round off to nearest integer) [Here $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x , y and z -axes respectively]

[25 Feb, 2021 (Shift-I)]

128. The electric field in a region is given by

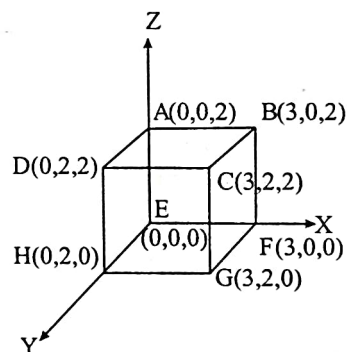
$$\vec{E} = \frac{2}{5} E_0 \hat{i} + \frac{3}{5} E_0 \hat{j} \text{ with } E_0 = 4.0 \times 10^3 \frac{N}{C}$$

The flux of this field through a rectangular surface area 0.4 m^2 parallel to the $Y-Z$ plane is Nm^2C^{-1}

[17 March, 2021 (Shift-II)]

129. An electric field $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j} \text{ N/C}$ passes through the box shown in figure. The flux of the electric field through surface $ABCD$ and $BCGF$ and marked as ϕ_1 and ϕ_2 respectively. The difference between $(\phi_1 - \phi_2)$ is (in Nm^2/C)

[9 Jan, 2020 (Shift-II)]



ANSWER KEY

- | | | | | | | | | | |
|------------|------------|------------|-----------|------------|----------|-----------|------------|------------|----------|
| 1. (*) | 2. (d) | 3. (b) | 4. [3] | 5. (2) | 6. [3] | 7. [3] | 8. (a) | 9. (2) | 10. (a) |
| 11. (b) | 12. [5440] | 13. [300] | 14. [2] | 15. [24] | 16. (a) | 17. (b) | 18. (d) | 19. (a) | 20. (b) |
| 21. (d) | 22. [17] | 23. (b) | 24. (c) | 25. (d) | 26. (c) | 27. [20] | 28. [6000] | 29. [36] | 30. (a) |
| 31. [12] | 32. (c) | 33. (c) | 34. (a) | 35. (b) | 36. (c) | 37. (b,d) | 38. (b) | 39. (c) | 40. (a) |
| 41. (a) | 42. (c) | 43. (b) | 44. (c) | 45. [5] | 46. (a) | 47. (b) | 48. [6] | 49. (b) | 50. (c) |
| 51. (a) | 52. (a) | 53. (b) | 54. (a) | 55. (a) | 56. (a) | 57. (c) | 58. (d) | 59. (a) | 60. (a) |
| 61. (a) | 62. (b) | 63. (d) | 64. (b) | 65. (d) | 66. (a) | 67. (a) | 68. (a) | 69. (d) | 70. [12] |
| 71. (c) | 72. (c) | 73. [2.00] | 74. [16] | 75. (c) | 76. (b) | 77. (*) | 78. (a) | 79. [6] | 80. (d) |
| 81. (d) | 82. (a) | 83. (d) | 84. (c) | 85. (a) | 86. (b) | 87. (b) | 88. [8] | 89. [45] | 90. [2] |
| 91. [8] | 92. (b) | 93. (b) | 94. (a) | 95. (d) | 96. (a) | 97. [1] | 98. (c) | 99. (a) | 100. (d) |
| 101. (b) | 102. (d) | 103. [15] | 104. (b) | 105. (b) | 106. (d) | 107. [48] | 108. (d) | 109. (c) | 110. (a) |
| 111. (c) | 112. (b) | 113. (2) | 114. (16) | 115. (b) | 116. (2) | 117. [12] | 118. (b) | 119. (16) | 120. (d) |
| 121. [640] | 122. (*) | 123. (c) | 124. (a) | 125. [226] | 126. [4] | 127. [1] | 128. [640] | 129. [-48] | |